AARON ANGELO CRISOLOGO

Computer Science Graduate | Full-Stack Developer

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PROFILE

Computer Science graduate with with a strong focus on cross-platform mobile application development, modern web development, full-stack engineering, and artificial intelligence. Demonstrated ability to deliver production-viable mobile experiences, intuitive user interfaces, and robust state management architectures. Hands-on industry experience building enterprise mobile apps utilizing Ionic and Angular. Thesis on lightweight SLMs for low-resource Philippine languages demonstrates the ability to ship production-viable ML systems under tight hardware constraints. Adept at leveraging cloud platforms (GCP) and implementing secure network architectures. Seeking to apply combined software engineering skills in a collaborative environment focused on technological innovation.

EDUCATION

Bachelor of Science in Computer Science — Batangas State University-TNEU, Alangilan

July 2026 | GWA: 1.69 / 5.0

COURSEWORK

Machine Learning · Web Development · Fundamentals of Data Science · Advanced Software Engineering · IoT · Technopreneurship · Network Security & Information Assurance · Cloud Computing & Infrastructure

FOCUS Web Development · Full-Stack Mobile Development · AI/ML Engineering · Cybersecurity

TECHNICAL SKILLS

Languages	Python, Java, C++, C#, JavaScript, TypeScript, Dart, SQL
Web / Mobile	React, Angular, Ionic, Flutter, .NET, Node.js, Laravel, HTML/CSS, Tailwind CSS
AI/ML & Data	PyTorch, YOLOv8, QLoRA, Knowledge Distillation, BERTScore, Google Colab, NumPy
Cloud / Infra	Google Cloud Platform, Network Configuration, Git
Certifications	CCNA: Introduction to Networks — Cisco (2023) | ISC2 Candidate (2024) | Google Cloud Computing Foundations (2024) | Build a Secure Google Cloud Network (2024) | Prepare Data for ML APIs on Google Cloud (2024)

PROJECT EXPERIENCE

Lightweight SLM for Cebuano & Ilocano QA on Edge Devices (Thesis)

July–May 2026 | Gemma 3 4B · QLoRA (4-bit) · Knowledge Distillation · PyTorch · BERTScore · LM Studio

- Fine-tuned a Gemma 3 4B SLM on 30,000 QA pairs for two low-resource Philippine languages (Cebuano, Iloko) using QLoRA 4-bit precision; achieved under 6 GB RAM footprint, enabling offline inference on consumer devices.
- Generated synthetic training data via knowledge distillation from larger models with human validation for linguistic accuracy.
- Outperformed base model by +9.0% average BERTScore F1 across all target languages; all gains statistically significant ($p < 0.001$).
- Benchmarked on real consumer mobile hardware, confirming sub-6 GB RAM usage and practical decode throughput.

X-Bsky-Mirror — Automated Cross-Platform Scraping and Posting Bot

2026 | Python · Node.js · Puppeteer · atproto · GitHub Actions · ffmpeg

- Automated mirroring bot that scrapes tweets (text, images, and videos) from a target Twitter/X account using Puppeteer with stealth plugins to avoid bot detection, then reposts them to Bluesky via the atproto API.
- Handles the full media pipeline: downloads images in high resolution, processes video m3u8 manifests with ffmpeg, and generates OpenGraph link-card previews for text-only posts.
- Implements deduplication by checking the last 2 Bluesky posts before publishing, and converts Twitter URLs, hashtags, and mentions to Bluesky rich-text clickable links.
- Deployed as a GitHub Actions cron job (runs twice daily at 5:15 AM and 10:10 PM UTC); includes simulation mode for local testing without live Twitter access.

Computer Vision-Based Adaptive Traffic Light System

Aug–Dec 2025 | YOLOv8 Nano · Deep Q-Network (DQN) · PyTorch · Arduino · Python · OpenCV

- Fine-tuned YOLOv8 Nano on a custom dataset of Philippine road scenes; detects 5 object classes (cars, motorcycles, PUVs, pedestrians, lanes) with mAP50 of 0.683.
- Designed a DQN reinforcement-learning module that dynamically selects green-light duration (5–25 s) per axis based on real-time vehicle counts.
- Integrated CV pipeline with Arduino prototype controlling 4 LED traffic modules and pedestrian inputs; implemented 3-layer safety override logic.
- Built entire prototype on ~Php 1,400 hardware budget using Google Colab.

Class Compass — University Companion App

2024 | Flutter · Dart · Node.js · Express · MySQL

- Cross-platform mobile app (Flutter/Dart) with a Node.js/Express backend consolidating class cancellations, emergency announcements, academic calendars, and schedule management into one interface for BatStateU students.
- Implemented department-detection login so each user receives personalised announcements and updates relevant to their specific academic environment.
- Built schedule manager with cancellation tracking, announcements system with admin moderation, and a resource hub with emergency contacts.
- Achieved 92% user satisfaction in pilot testing; team of 4 developers using Git for collaborative version control.

CPU Scheduling Algorithm Visualizer

2024 | JavaScript · HTML5 Canvas · CSS

- Browser-only learning platform (zero install, no backend) simulating 6 CPU scheduling algorithms: FCFS, SJF, SRTF, Non-Preemptive Priority, Preemptive Priority, and Round-Robin.
- Rendered real-time animated Gantt charts via HTML5 Canvas API; auto-calculates Average Waiting Time and Average Turnaround Time per run.
- Built a side-by-side algorithm comparison mode and step-by-step interactive tutorials explaining core scheduling concepts.
- Architected with modular JavaScript (one file per algorithm) for clean extensibility; responsive across desktop and mobile browsers.

INDUSTRY EXPERIENCE

Software Engineering Intern — OJT at ICT Services Central Batangas City

Batangas, Philippines

- Developed production mobile apps using Angular and Ionic, building UI components from Figma designs integrated with Laravel backends.
- Implemented Google Workspace OAuth 2.0 SSO via Laravel Socialite + Sanctum for token-based API auth, restricting access by Hosted Domain.
- Built SSC Payments UI and Performance Evaluation module for a student mobile app, including dynamic data rendering, validation, progress tracking, and custom modals.
- Created reusable component templates adopted team-wide; global styling with custom animations (crossfade, scroll-to-top).
- Built CRUD features in Laravel for an employee portal; collaborated on backend design for the OEAMS module.
- Used Git for team collaboration (feature branches, merges) and ADB for on-device testing.

ADDITIONAL INFORMATION

Languages	English (Fluent) · Filipino (Native)
Interests	Open-source contribution · Cloud architecture · NLP for low-resource languages · Embedded systems · IoT
Soft Skills	Team Collaboration · Problem-Solving · Technical Writing · Agile Workflows
Professional Dev.	Regularly complete online courses in emerging technologies (ML, cloud security, full-stack dev) and maintain an active GitHub portfolio.